

Zero-Shot Industrial Anomaly Detection via Spatial Bottleneck Adaptation and Adaptive Tiling

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<https://github.com/yongheng0011/ASCLIP>

Abstract

We propose a zero-shot industrial anomaly detection method for the VAND 4.0 @ CVPR 2026 Challenge, built on AdaptCLIP with two improvements to the visual adapter. First, we introduce a Spatial Bottleneck Adapter (SBA) that replaces the patch-level residual MLP with a depthwise 3×3 convolution, enabling local spatial interaction among neighboring patch tokens. Second, we replace bilinear interpolation with a learnable convolutional decoder for progressive, spatially coherent upsampling of the anomaly similarity map. For inference on high-resolution MVTec AD 2 images, we apply an adaptive tiling strategy (518×518 tiles, overlap-averaged boundaries) and a statistics-based thresholding scheme requiring no validation data. Trained solely on MVTec AD without fine-tuning or test-time augmentation, our method achieves mean AucPro_{0.05} / SegF1 / ClassF1 of 28.54% / 24.55% / 70.72% under standard lighting and 28.69% / 22.84% / 71.13% under mixed lighting across all eight categories, demonstrating competitive zero-shot generalisation to real-world industrial inspection scenarios.

1. Introduction

1.1. Background

Industrial anomaly detection is a critical component of automated quality inspection in modern manufacturing. The goal is to identify defective or abnormal items in product images, typically without access to labeled anomalous samples during training. This unsupervised or one-class learning paradigm has been extensively studied with datasets such as MVTec AD [1] and VisA [7], yet adapting trained models to new object categories or environmental conditions remains a fundamental challenge.

Recent advances in vision-language models, particularly CLIP [5], have opened a new avenue for zero-shot anomaly detection. Methods such as WinCLIP [4], AnomalyCLIP [6], and AdaptCLIP [2] demonstrate that CLIP’s rich visual-semantic representations can be leveraged to detect anomalies without category-specific training data. AdaptCLIP, in particular, proposes three lightweight adapters—visual, textual, and prompt-query—that enable zero-shot and few-shot anomaly detection across diverse domains while preserving CLIP’s original representational ability.

1.2. Challenge Description

The VAND 4.0 @ CVPR 2026 Challenge Industrial Track adopts the MVTec AD 2 dataset [3], which significantly advances the realism of industrial anomaly detection benchmarks. MVTec AD 2 includes eight challenging real-world scenarios with high-resolution images captured under varying lighting conditions. Compared to its predecessor, it introduces transparent and overlapping objects, dark-field and backlight illumination, and extreme intra-class variance, posing substantial challenges for existing methods.

The Zero-Shot Sub-Track requires participants to develop models that do not use any MVTec AD 2 images during training—neither normal nor anomalous—while still achieving reliable pixel-level anomaly segmentation (SegF1) on both standard (private) and mixed-lighting (private_mixed) test sets. Since anomaly maps are typically continuous, this evaluation also requires selecting a single decision threshold—a challenge that the scientific community has often not considered, yet is essential for real-world application deployment.

Our entry addresses these challenges by adapting AdaptCLIP for the zero-shot industrial setting, introducing spatial awareness into the visual adapter and designing a tiling-based inference strategy for large, high-resolution images.

2. Methodology

2.1. Model Design

Our method is built on the AdaptCLIP framework [2], which extends CLIP with three pluggable adapter modules: a Visual Adapter, a Textual Adapter, and a Prompt-Query Adapter. In the zero-shot setting, we employ only the visual and textual adapters, as no reference normal images are provided at inference time. The model is pre-trained on MVTec AD [1] and applied directly to MVTec AD 2 [3] without any fine-tuning.

2.1.1. Visual-S Adapter

The original Visual Adapter in AdaptCLIP applies a residual MLP (ResMLP) independently to both the global image token and the local patch tokens, followed by bilinear interpolation to upsample the cosine similarity map to the input resolution. While sufficient for global-level adaptation, the per-token MLP applied to patch features is spatially unaware: it treats each patch independently and cannot capture local structural patterns among neighboring patches, such as edge continuity or texture gradients. Furthermore, bilinear interpolation of the low-resolution similarity map (37×37 for ViT-L/14@336px with 518×518 input) discards fine-grained spatial details that could be recovered through learned upsampling.

We retain the ResMLP for the global image token without modification, as global-level anomaly classification does not require spatial context. However, we replace the patch-level pipeline with two components: a **Spatial Bottleneck Adapter** for richer local feature interaction, and a **Similarity Map Decoder** for learnable upsampling.

Global Branch. The global image token $f^q \in \mathbb{R}^D$ is adapted using the original two-layer residual MLP:

$$f^{q'} = f^q + \text{MLP}(f^q; \theta_v^q). \quad (1)$$

where θ_v^q is learnable parameters. This branch is unchanged from the original AdaptCLIP design.

Local Branch – Spatial Bottleneck Adapter. Given the sequence of patch tokens of shape $(B, 1+N, D)$, where $N = H \times W$ and D is the feature dimension, the Spatial Bottleneck Adapter (SBA) operates exclusively on the N patch tokens (excluding the CLS token) as follows:

- **Channel reduction:** a linear projection maps D -dimensional tokens to D/r -dimensional tokens ($r=4$ by default).
- **Spatial interaction:** patch tokens are reshaped into a 2D spatial grid of shape $(B, D/r, H, W)$ and processed with a depthwise 3×3 convolution (groups= D/r , padding=1) to aggregate local spatial context among neighboring patches, followed by a GELU activation.
- **Channel expansion:** tokens are flattened back to $(B, N, D/r)$ and projected back to D dimensions via a linear layer, followed by dropout ($p=0.1$).
- **Residual connection:** the adapter output is added to the input patch tokens:

$$F_i^{q'} = F_i^q + \text{SBA}(F_i^q; \theta_v^l) \quad (2)$$

where $F_i^q \in \mathbb{R}^{B \times N \times D}$ denotes the patch token feature matrix output by the i -th layer; $F_i^{q'}$ is the adapted patch feature after the SBA residual connection; and θ_v^l represents the learnable parameters of the SBA, comprising the weights of the channel reduction/expansion linear projections and the depthwise convolutional kernel.

The adapted patch features are then ℓ_2 -normalized and used to compute a low-resolution similarity map against the static textual embeddings via cosine similarity with temperature $\tau=0.07$, followed by softmax over the two classes to yield a soft assignment map of shape $(B, 2, H, W)$.

Local Branch – Similarity Map Decoder. Instead of bilinear interpolation, we upsample the low-resolution similarity map $(B, 2, 37, 37)$ to the full input resolution $(B, 2, 518, 518)$ using a learnable convolutional decoder with three progressive upsampling stages:

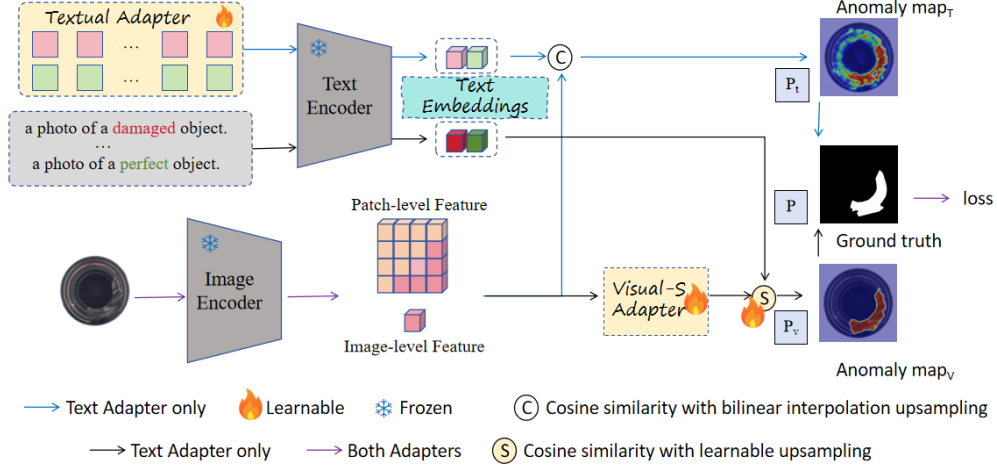


Figure 1. This is a structural diagram of our proposed method. We follow AdaptCLIP’s alternating training strategy and text adapter, replace the visual adapter with the Visual-S Adapter, and replace the approach of upsampling the similarity map to the input image size after similarity computation during visual adapter training with a learnable upsampling module.

- **Stage 1** (37→74): TransposedConv2d($2 \rightarrow C$, $k=4$, $s=2$, $p=1$) + BN + ReLU + Conv2d($C \rightarrow C$, $k=3$, $p=1$) + BN + ReLU.
- **Stage 2** (74→148): TransposedConv2d($C \rightarrow C/2$, $k=4$, $s=2$, $p=1$) + BN + ReLU + Conv2d($C/2 \rightarrow C/2$, $k=3$, $p=1$) + BN + ReLU.
- **Stage 3** (148→296): TransposedConv2d($C/2 \rightarrow C/4$, $k=4$, $s=2$, $p=1$) + BN + ReLU + Conv2d($C/4 \rightarrow 2$, $k=1$).
- **Final resize**: bilinear interpolation from 296×296 to 518×518 .

Here $C=32$ is the mid-channel dimension, and all convolution weights are initialized with Kaiming normal initialization. A final softmax is applied over the two-class channel dimension to produce the pixel-level anomaly map. This design enables the model to learn spatially adaptive upsampling patterns—such as sharpening anomaly boundaries—rather than relying on fixed bilinear kernels.

This two-part local branch design enables the adapter to capture both local spatial correlations among patch tokens (via SBA) and spatially coherent anomaly localization at full resolution (via the learned decoder), while introducing only minimal additional parameters (one depthwise convolution layer and a small transposed-convolution decoder).

2.1.2. Textual Adapter

We retain the original Textual Adapter from AdaptCLIP without modification. It learns two-class prompt embeddings (normal and abnormal) fed into the frozen CLIP textual encoder. The length of learnable prompt embeddings is set to 12. This adapter is trained jointly with the visual adapter on the base training datasets using alter-

nating optimization.

2.1.3. No Image Augmentation

Unlike some anomaly detection methods that apply aggressive data augmentation during inference or preprocessing (e.g., test-time color jitter or geometric transforms), our method applies no image augmentation at any stage. Images are processed in their original colorimetry, preserving the lighting-related features that are critical for distinguishing normal from anomalous regions under the varied illumination conditions of MVTec AD 2.

2.1.4. Other Settings

All experiments were conducted on a server equipped with an NVIDIA GeForce RTX 4090 D GPU, with the random seed fixed at 10 to ensure reproducibility. The Adam optimizer was used with a learning rate of 0.001 and momentum parameters (0.5, 0.999); the model was trained for 10 epochs with a batch size of 8.

2.2. Dataset & Evaluation

2.2.1. Data Utilization

In strict adherence to the Zero-Shot Sub-Track requirements, our model is not trained or fine-tuned on any images from the MVTec AD 2 dataset [3]. Pre-training follows the protocol of AdaptCLIP [2]: the model is trained on the test split of MVTec AD [1] and evaluated zero-shot on target datasets.

2.2.2. Adaptive Tiling for High-Resolution Images

MVTec AD 2 contains high-resolution images with non-square aspect ratios, which exceed the 518×518 input resolution used by AdaptCLIP. To handle this, we apply an adaptive tiling strategy:

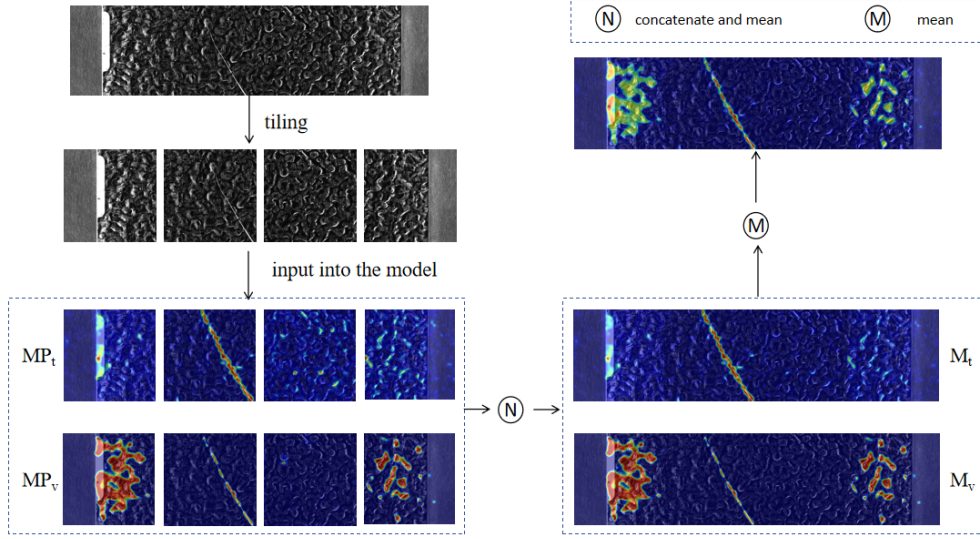


Figure 2. This is a structural diagram of the inference process of our proposed method. After patch partitioning, the patch-level anomaly maps MP generated independently by each branch are first stitched together, with overlapping regions averaged arithmetically; the anomaly maps M obtained from the two branches are then arithmetically averaged to produce the final anomaly map.

- The short edge of the image is rescaled to 518 pixels; the long edge is scaled proportionally.
- Tiles of size 518×518 are extracted by sliding along the long axis with a stride of 518 pixels (no forced overlap).
- If the final tile would fall outside the image boundary, it is fixed to the last 518 pixels of the long axis (i.e., the tile may overlap with the previous one). The overlapping region’s anomaly scores are resolved by arithmetic averaging.

After each branch independently generates its patch-level anomaly maps MP , the tile maps are stitched together (overlapping regions averaged), and the maps from the textual and visual branches are arithmetically averaged to produce the final anomaly map M (see Figure 2).

This strategy ensures full coverage of the image without information loss at boundaries, while keeping computational cost proportional to image area.

2.2.3. Thresholding Strategy

Converting continuous anomaly maps to binary segmentation masks requires selecting a threshold. Instead of using a validation set, we compute a single threshold per object category by pooling anomaly scores across all test subfolders of that category (e.g., `test_private` and `test_private_mixed` are treated jointly for the *can* class). For a given category, the threshold is set to the smallest integer n such that $(\mu + n \cdot \sigma) > 0.5$, where μ and σ are the mean and standard deviation of

the anomaly scores computed over all images in that category. Formally:

$$\text{threshold} = \min\{\mu + n \cdot \sigma \mid n \in \mathbb{Z}_{\geq 0}, \mu + n \cdot \sigma > 0.5\}. \quad (3)$$

This ensures the threshold always exceeds the 0.5 decision boundary, avoiding degenerate cases where the entire image is predicted as normal. The integer constraint on n provides a coarse but stable discretization that avoids overfitting to the score distribution of any particular subfolder.

2.3. Evaluation Criteria

The VAND 4.0 @ CVPR 2026 Challenge evaluates submissions on three complementary metrics, all computed on the binarized anomaly maps using the per-category threshold described in Sec. 2.2.3.

AucPro_0.05. The area under the per-region-overlap (PRO) curve, integrated up to a false positive rate (FPR) of 0.05 [3]. Formally, for a test set of $|\mathcal{D}_{\text{test}}|$ images each containing k_i connected anomalous components $\{C_{i,j}\}$, the PRO is defined as

$$\text{PRO} = \frac{1}{k} \sum_{i=1}^{|\mathcal{D}_{\text{test}}|} \sum_{j=1}^{k_i} \frac{|P_{\text{ano}} \cap C_{i,j}|}{|C_{i,j}|}, \quad (4)$$

where P_{ano} denotes the set of pixels predicted as anomalous for a given threshold. Unlike pixel-level AU-ROC, PRO weights each connected defect region equally regardless of size, which is critical for datasets containing

both large and extremely small defects. The integration limit is set to 0.05 (rather than the conventional 0.30) to encourage precise localization, since MVTec AD 2 [3] contains many small defects in large high-resolution images.

SegF1. The pixel-level F_1 score on binarized anomaly maps:

$$\text{SegF1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (5)$$

where precision and recall are computed by comparing the thresholded anomaly map pixel-by-pixel against the ground-truth segmentation mask. This metric captures segmentation quality after thresholding and is thus sensitive to the choice of threshold.

ClassF1. The image-level classification F_1 score derived from the same segmentation threshold. An image is classified as *anomalous* if at least one pixel exceeds the threshold; otherwise it is classified as *normal*. ClassF1 is then computed as the harmonic mean of image-level precision and recall. This metric measures practical reject/accept decision quality under the same threshold used for segmentation, reflecting real-world deployment requirements.

3. Results

3.1. Performance Metrics

Table 1 reports per-category and mean results on the two private test sets. Our method achieves mean AucPro_0.05 / SegF1 / ClassF1 of **28.54% / 24.55% / 70.72%** on $\text{TEST}_{\text{priv}}$ and **28.69% / 22.84% / 71.13%** on $\text{TEST}_{\text{priv,mix}}$, demonstrating stable generalisation under both standard and mixed lighting conditions.

Performance varies markedly across object categories, reflecting the heterogeneous difficulty of MVTec AD 2. Rice, Walnuts, and Sheet Metal achieve the highest SegF1 scores (46.20%, 40.57%, and 43.06% on $\text{TEST}_{\text{priv}}$), benefiting from spatially repetitive textures and distributed anomaly patterns that are well-suited to CLIP’s patch-level representations. Fruit Jelly and Vial show moderate localisation quality (24.54% and 21.01% SegF1), despite the additional challenges of transparency and backlit illumination that complicate the definition of a normal appearance.

In contrast, Can and Wall Plugs exhibit substantially lower SegF1 scores (0.60% and 0.42% on $\text{TEST}_{\text{priv}}$). Can images feature highly reflective metallic surfaces whose specular highlights vary strongly with lighting angle, producing anomaly score distributions that are difficult to threshold reliably. Wall Plugs contain many small, overlapping objects with fine geometric detail; in the zero-shot setting, without target-domain normal prototypes, localising individual plug defects among dense clutter is inherently challenging.

Table 1. Results on $\text{TEST}_{\text{priv}}$ / $\text{TEST}_{\text{priv,mix}}$.

Object	AucPro_0.05 (%)	SegF1 (%)	ClassF1 (%)
Can	5.93 / 5.87	0.60 / 0.58	56.45 / 59.25
Fabric	15.12 / 15.50	19.97 / 18.69	71.30 / 72.13
Fruit Jelly	49.43 / 49.88	24.54 / 24.93	83.88 / 84.16
Rice	25.18 / 27.83	46.20 / 46.38	31.53 / 31.96
Sheet Metal	41.88 / 36.76	43.06 / 31.54	87.13 / 86.53
Vial	39.16 / 39.28	21.01 / 20.68	83.05 / 83.05
Wall Plugs	5.96 / 6.14	0.42 / 0.44	75.55 / 75.55
Walnuts	45.63 / 48.25	40.57 / 39.48	76.87 / 76.45
Mean	28.54 / 28.69	24.55 / 22.84	70.72 / 71.13

Notably, ClassF1 scores are substantially higher than SegF1 across all categories (mean 70.72% vs. 24.55% on $\text{TEST}_{\text{priv}}$), confirming that the statistical thresholding scheme (Sec. 2.2.3) reliably distinguishes anomalous from normal images even when pixel-level boundary precision is limited. This reflects an important practical property: the method supports reliable image-level accept/reject decisions in zero-shot deployment where no validation data is available for threshold calibration.

Comparing $\text{TEST}_{\text{priv}}$ and $\text{TEST}_{\text{priv,mix}}$, performance is largely consistent across all categories—the mean SegF1 decreases by only 1.71 percentage points under mixed lighting. This robustness can be attributed to CLIP’s rich multi-modal representations, which encode photometric invariance from large-scale diverse pre-training, combined with the deliberate decision to avoid illumination normalisation at inference time (Sec. 2.1.3).

3.2. Ablation Study

To validate the individual contribution of each proposed component, we conduct ablation experiments on TEST_{pub} by incrementally enabling one component at a time. Three components are examined: the **Spatial Bottleneck Adapter (SBA)**, which replaces the original patch-level ResMLP with a depthwise-convolution bottleneck; the **Similarity Map Decoder (SMD)**, which replaces bilinear interpolation with a learnable transposed-convolution upsampler; and **Adaptive Tiling (AT)**, which handles high-resolution inputs via sliding-window inference instead of direct rescaling to 518×518 . Results are reported in Table 2 as mean SegF1 over all eight categories.

The baseline configuration (ResMLP + bilinear interpolation + direct resize, all components disabled) achieves 16.67% SegF1. Enabling SBA alone improves this to 17.32% (+0.65 pp), demonstrating that introducing local spatial context into patch token adaptation yields a consistent gain even in isolation. Adding

Table 2. Ablation study on TEST_{pub} (mean over all eight categories). ✓ = component enabled; ✗ = replaced by its baseline counterpart (ResMLP / bilinear interpolation / direct resize).

SBA	SMD	AT	SegF1 (%)
✗	✗	✗	16.67
✓	✗	✗	17.32
✓	✓	✗	19.46
✓	✓	✓	21.53

SMD raises performance to 19.46% (+2.14 pp), demonstrating that learnable progressive upsampling recovers spatial detail at anomaly boundaries that fixed bilinear kernels discard. Finally, enabling AT further improves SegF1 to 21.53% (+2.07 pp), confirming that tiling-based inference preserves fine-grained defect information in high-resolution images that would otherwise be lost under aggressive downsampling.

The gains from all three components are complementary and individually significant: SBA addresses local feature interaction, SMD handles spatially coherent upsampling, and AT ensures full-image coverage at native resolution. Together, they achieve a cumulative improvement of +4.86 pp SegF1 over the all-disabled baseline, validating the motivation for each design choice.

3.3. Qualitative Results

Figure 2 illustrates the full inference pipeline on a representative Sheet Metal example from TEST_{pub} , showing how anomaly maps from the textual and visual branches are tiled, merged, and arithmetically averaged to produce the final prediction. The Visual-S Adapter produces high-contrast activations over the anomalous scratch region while maintaining low background responses over defect-free surface areas, demonstrating the effectiveness of spatial bottleneck feature adaptation.

Across categories, localisation quality correlates with the texture regularity of the object. For Walnuts and Rice, where anomalies manifest as local deviations from a statistically uniform background, the learned patch-level representations yield compact and well-centred anomaly regions. For Sheet Metal, the adaptive tiling strategy is especially beneficial: thin linear defects (*e.g.*, scratches) that would occupy only a handful of pixels after full-image downsampling are preserved at native resolution within each tile, resulting in clearly detected elongated anomaly maps.

For transparent categories (Vial, Fruit Jelly), CLIP’s semantic prior provides a useful starting point, but score distributions are more diffuse, reflecting the inherent ambiguity between normal optical effects (*e.g.*, refraction, caustics) and actual material defects. Can and Wall

Plugs remain the most challenging: the former due to extreme specular variation across lighting conditions, the latter due to dense spatial clutter that yields ambiguous patch-level responses. These observations motivate future work on few-shot or target-domain adaptation strategies for categories that fall outside CLIP’s zero-shot discriminative capacity.

4. Discussion

4.1. Challenges & Solutions

Zero-shot anomaly detection on MVTec AD 2 [3] presents several unique challenges beyond those in standard benchmarks:

- **Varied illumination:** Images are captured under at least four different lighting conditions per scene. Our method does not apply any illumination normalization, relying instead on CLIP’s robust multi-modal representations trained on diverse internet images. The alternating learning strategy of Adapt-CLIP [2] further ensures that neither the visual nor textual adapter is over-adapted to a specific illumination regime.
- **High-resolution images:** MVTec AD 2 images are significantly larger than those in MVTec AD [1]. Direct resizing to 518×518 would lose fine-grained defect details. Our tiling strategy mitigates this by processing full-resolution local patches and averaging overlapping predictions.
- **Single-threshold constraint:** Unlike image-adaptive thresholding, the challenge requires a single threshold per category. Our mean $+n \times \text{std}$ rule with the constraint threshold > 0.5 provides a statistically grounded, data-driven threshold that is robust across categories without requiring manual tuning.
- **Transparent and overlapping objects:** Some MVTec AD 2 categories (*e.g.*, vial, fruit jelly) include transparent or semi-transparent objects. CLIP’s rich semantic features, trained on large-scale web imagery, provide a strong prior for distinguishing normal transparency from anomalous regions.

4.2. Model Robustness & Adaptability

A key strength of our approach is that it operates without any target-domain fine-tuning. The Spatial Bottleneck Adapter is trained on source datasets (MVTec AD / VisA [7]) and is applied directly to MVTec AD 2, yet benefits from the structural inductive bias of spatial convolution that generalizes across different object types. The depthwise convolution captures local spatial patterns—such as edge discontinuities and texture irregularities—that are domain-agnostic indicators of anomalous regions.

The choice to omit image augmentation during inference is deliberate: MVTec AD 2’s lighting variation is a test-time distribution shift, not a training-time one. Applying random color jitter or brightness changes during inference could corrupt the illumination-specific normal features that CLIP has learned to represent, potentially degrading detection performance on consistent-lighting categories.

4.3. Future Work

Several directions could further improve our method:

- **Adaptive spatial resolution:** The current tiling strategy uses a fixed tile size of 518×518 . A multi-scale tiling approach with tiles of different sizes could better handle categories with both macro- and micro-scale defects.
- **Prompt engineering:** The current textual adapter uses generic normal/abnormal prompts. Incorporating category-specific visual descriptions—without violating the zero-shot constraint—could enhance discrimination for specialized categories like sheet metal or circuit components.
- **Computational efficiency:** The Spatial Bottleneck Adapter adds minimal parameters but the tiling strategy increases inference time linearly with the number of tiles. Future work could explore efficient attention-based spatial pooling to handle large images within a single forward pass.

5. Conclusion

We presented a zero-shot industrial anomaly detection method for the VAND 4.0 @ CVPR 2026 Challenge Industrial Track. By augmenting AdaptCLIP’s visual adapter with a *Spatial Bottleneck Adapter*, we introduce depthwise spatial convolution into patch-level feature adaptation, enabling richer local feature interaction among neighbouring patch tokens. Combined with a learnable *Similarity Map Decoder* for spatially coherent upsampling, an adaptive tiling strategy for high-resolution inference, and a principled statistical thresholding scheme, our method achieves mean AucPro.0.05 / SegF1 / ClassF1 of 28.54% / 24.55% / 70.72% on $\text{TEST}_{\text{priv}}$ and 28.69% / 22.84% / 71.13% on $\text{TEST}_{\text{priv,mix}}$ across all eight MVTec AD 2 categories. Ablation studies confirm that each of the three proposed components contributes independently to overall segmentation performance, with a combined gain of +4.86 pp SegF1 over the all-disabled baseline on TEST_{pub} . These results demonstrate competitive zero-shot generalisation to real-world industrial inspection scenarios—without any fine-tuning on target data, image augmentation, or category-specific engineering.

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